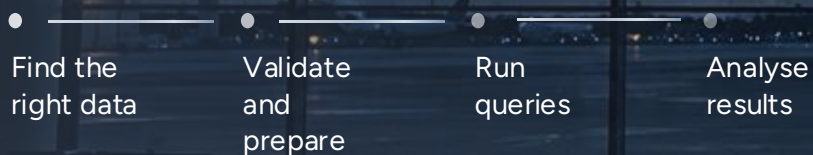


09:30

Revenue analysis for February 2026

I need the answer for the 10:00 meeting

09:30 the question lands



🕒 10:00 Management meeting

10:30 is already too late

From Dashboards to Decisions:

Building AI-Powered Analytics

A real-world journey from fragmented
operational data to business impact

Viktoriia Kniazeva
Solution Architect

HIGHER INSIGHTS
BRIGHTER JOURNEYS

AI adoption is moving faster than data readiness

88%

of organizations use AI in at least one business function

7%

say AI is fully scaled across the organization

26%

of CDOs are confident their data capabilities can support new AI-enabled revenue streams

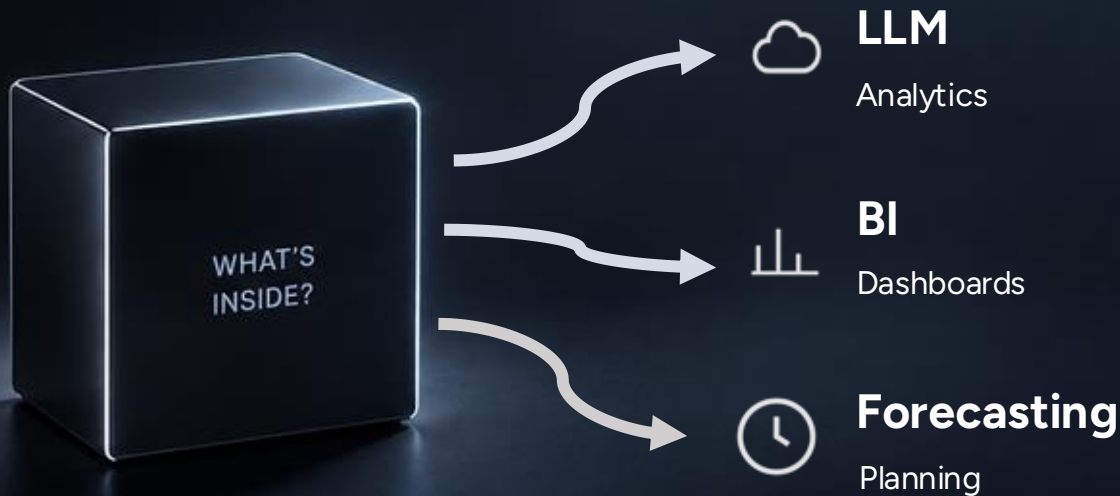
Sources: McKinsey, The state of AI ; IBM, Global Chief Data Officer Study



AI is scaling. Data foundations aren't keeping up.

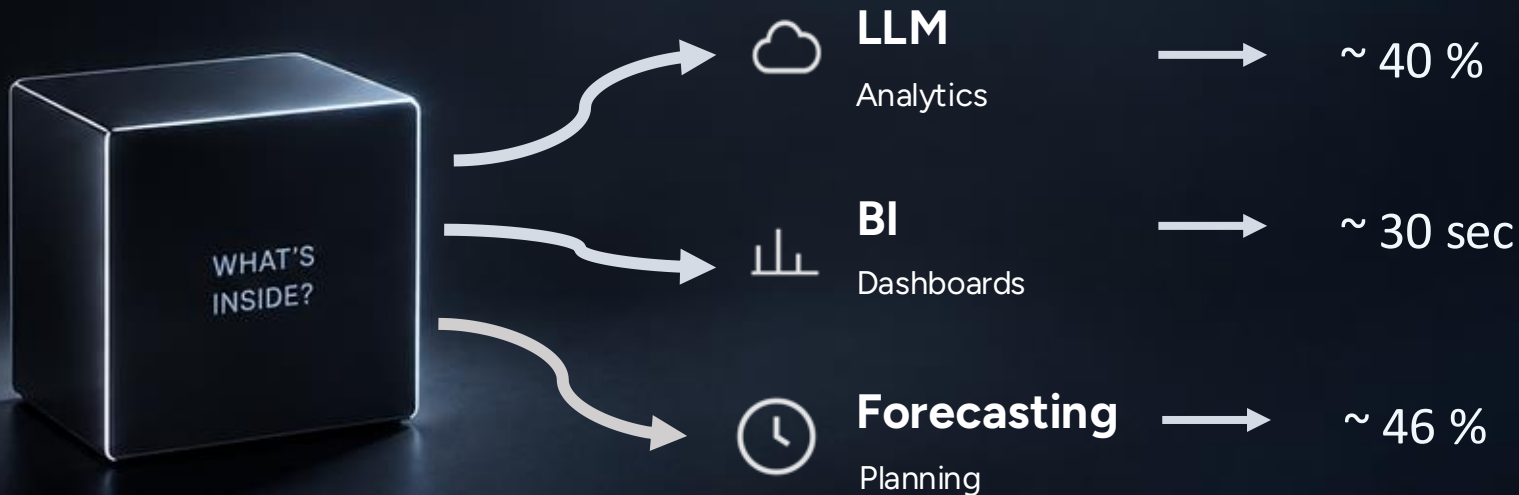
State 0:

AI was there.
The foundation wasn't.

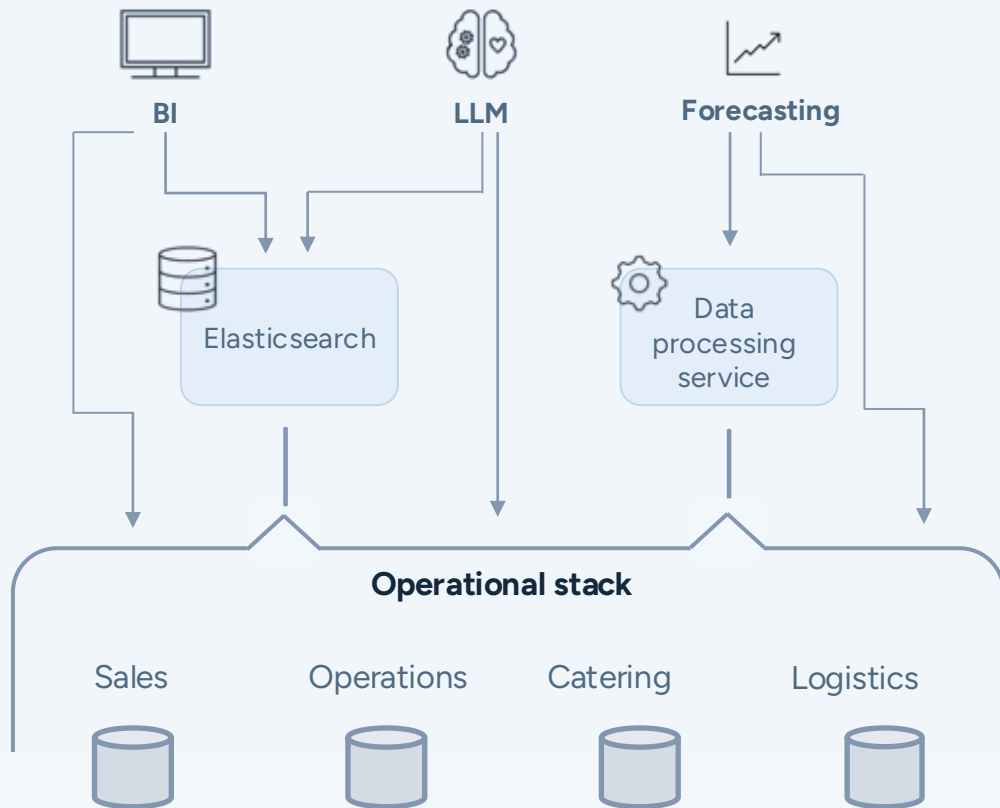


State 0:

AI was there.
The foundation wasn't.



Let's open the box.



-  **Direct DB access**
Operational data exposed to analytics
-  **Duplicated logic**
Business logic spread across services
-  **No semantic layer**
Same metric = same definition
-  **Manual pipelines**
Forecasting depends on custom scripts

What if the data could answer back?

Two questions. Real business decisions.

1

Revenue analysis in February
2026?



What if the data could answer back?

Two questions. Real business decisions.

1

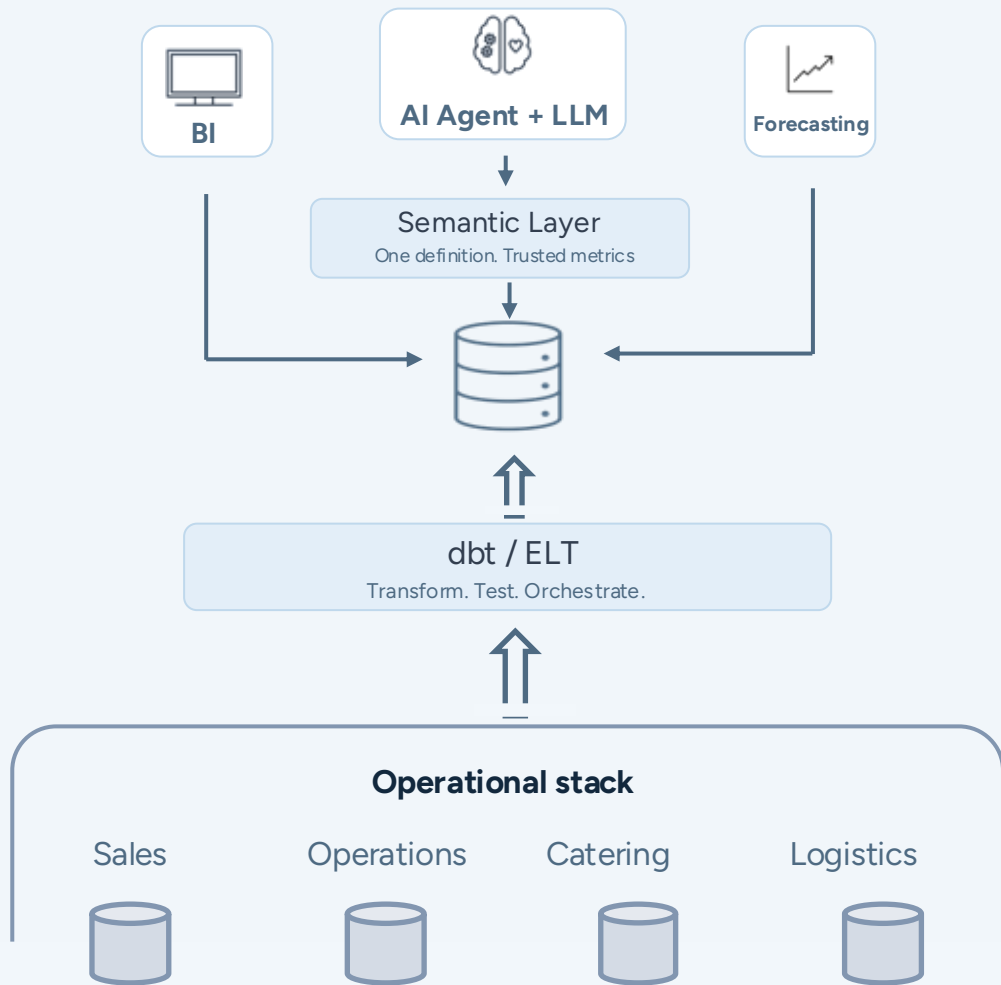
Revenue analysis in February 2026?



2

How many fresh sandwiches should we load tomorrow?





The architecture behind the experience

- 01 — **Centralise**
One analytical foundation
- 02 — **Standardise**
dbt + data quality
- 03 — **Govern**
Semantic layer + trusted metrics
- 04 — **Enable AI**
LLM and analytics over governed data

What actually happens when I ask a question?

08

What happened to sales in February?

AI Agent



Semantic Layer



Data Products



Analytical DWH



Operational Sources



USER

✗ The agent does NOT:

- ✗ Query operational databases directly
- ✗ Invent metric definitions
- ✗ Access arbitrary tables
- ✗ Execute unrestricted SQL

✓ The agent DOES:

- ✓ Resolve business intent
- ✓ Use governed metrics
- ✓ Generate constrained queries
- ✓ Return explainable, traceable results



01



Separate operational and analytical workloads

TRADE-OFF

Additional infrastructure and data pipelines

02



ELT with dbt

TRADE-OFF

Compute costs and modelling discipline across teams

03



Governed semantic layer

TRADE-OFF

Governance overhead and change management

04



AI over governed data

TRADE-OFF

Less flexibility for the agent compared to direct database access



Key architectural decisions

From data friction to decision speed



30s → 2s

Dashboard loading



92.3%

AI analytics accuracy



46% → 7%

Fresh-food waste



No manual fixes

Forecasting



So what actually made the difference?

01 Reliable data

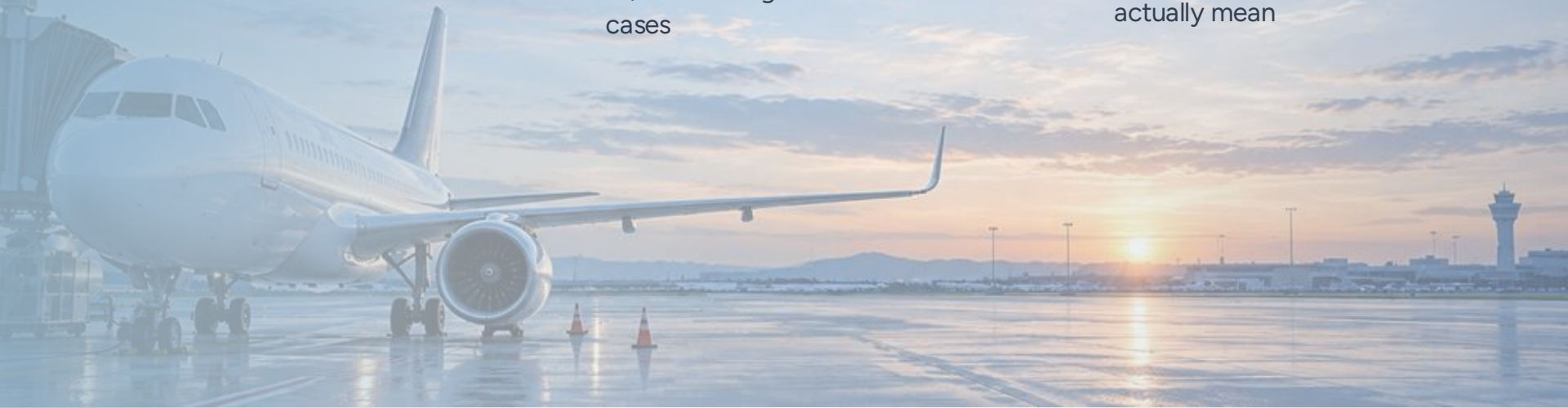
No AI can compensate for inconsistent data

02 Reusable data products

One foundation can power BI, AI, forecasting and future use cases

03 Governed meaning

AI needs to understand what "sales", "waste" and "revenue" actually mean



What does this mean for the business?



FASTER

Decisions happen in minutes, not hours



CHEAPER

Less manual analysis, less operational rework



SCALABLE

One data foundation can support new AI use cases

The architecture lesson

Don't build AI use cases one by one. Build the foundation that lets them scale.

Thanks!

Any Questions?



VIKTORIIA KNIAZEVA

SOLUTION ARCHITECT



LINKEDIN



GITHUB

